

Write a short note on MEMS.

MEMS (Micro-Electro-Mechanical Systems)

MEMS are very small devices that integrate **mechanical components, sensors, actuators, and electronics** on a single chip. They are fabricated using microfabrication technology and operate at a microscopic scale.

Components of MEMS:

- **Microsensors** – sense physical parameters like temperature, pressure, motion
- **Microactuators** – perform actions based on signals
- **Microelectronics** – process the data

Features:

- Small size and lightweight
- Low power consumption
- High precision and reliability
- Low cost in mass production

Applications:

- Smartphones (accelerometer, gyroscope)
- Automobiles (airbag sensors)
- Medical devices (biosensors)
- Industrial systems (pressure sensors)